Karl Ivan Alberto

+63 926 714 5410 | karlivan.c.alberto@gmail.com | linkedin.com/in/karlivanalberto | k14lb3.com

Education

Batangas State University — Malvar, Batangas
Bachelor of Science in Computer Science — *Sep. 2018 – Jul 2022*

Experience

Software Development Engineer — Sep 2022 – Present
Restoplus — *Adelaide, South Australia*

- Implemented new offline-first POS architecture enabling real-time peer-to-peer synchronization between POS terminals using IndexedDB, WebRTC, and Firebase Firestore.
- Integrated Uber Eats, including configuration, menu publishing, real-time item availability updates, and order ingestion into POS sales reporting.
- Integrated Tyro Payments, including configuration setup, terminal pairing, transaction logging, and end-of-day reporting with receipts.
- Integrated ANZ Worldline Payment Solutions, including MOTO transactions, split payments, partial refunds, transaction recovery, sales reporting, and daily reconciliation.
- Architected and implemented Google Ordering redirect migration from Google Food Ordering to Google Ordering Redirect.
- Architected and implemented type-safe Firebase callable Cloud Functions framework with middleware composition to improve reliability and maintainability.
- Designed and implemented Reorder feature, enabling customers to quickly recreate previous POS orders with ease.
- Designed and implemented centralized Item Summary generation logic and shared components used across POS, Admin Portal, Customer Portal, receipts, and email notifications.
- Modernized legacy POS Cart and Order History through UI redesign and refactoring, improving maintainability.
- Researched and prototyped search solutions using Vertex AI Search and Firebase queries for customer and order retrieval.
- Designed and implemented reusable React and SCSS components, including popouts, modals, and other core UI primitives, to ensure UI consistency across POS, Admin Portal, and Customer Portal.

Projects

95 | *TypeScript, React, Next.js, Zustand, StyleX, Biome* — Repository

- Windows 95 clone.

dotfiles | *Lua, zsh, tmux, Alacritty, Neovim, Lazygit* — Repository

- Terminal-based configuration for Unix systems optimized for development efficiency and speed.

Technical Skills

Languages: HTML, CSS, Sass/SCSS, JavaScript, TypeScript, Bash, Lua
Libraries & Frameworks: Express, Node.js, React, Next.js, Zustand, MobX, StyleX, Tailwind CSS, Jest, Cypress, Biome
Tools: Git, GitHub, Lazygit, Docker, Google Cloud Platform, Notion, Alacritty, Neovim, vim, tmux, zsh, OpenCode, Claude Code, Cursor CLI, Cursor